

Vinay Jampana

Senior Software Engineer – AI & Full-Stack

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Summary

Senior engineer with 5+ years of React/TypeScript, Node/NestJS and AWS experience who builds AI features end to end: order-agent pipelines composed on a visual canvas, product matching benchmarked on 15K production aliases, deterministic evals gated in CI, and OpenTelemetry traces, plus a team of five engineers that I lead.

Skills

AI / GenAI: LLM agents, tool calling, LangGraph, structured outputs (Zod), prompt engineering, LLM evaluation (ground-truth, hit@k), Langfuse, OpenTelemetry, OCR/vision pipelines, OpenSearch kNN retrieval, LiteLLM

Backend & Data: TypeScript, Node.js, NestJS, Python (FastAPI, LangGraph), PostgreSQL (pg_trgm), Prisma, Hasura/GraphQL, REST, KrakenD

Cloud & Infra: AWS (ECS, ECR, Lambda, S3, Amplify), Docker, GitHub Actions, New Relic

Frontend: React, TypeScript, Next.js, Redux Toolkit, Module Federation, Vitest

Engineering: Event-driven design, idempotency, feature flags and rollback, multi-tenant systems, RBAC, eval-driven development

Experience

Zotok.ai – Senior Software Engineer (AI & Full-Stack) May 2023 – Present

Hyderabad. B2B WhatsApp-commerce SaaS; customers include Sun Pharma, UltraTech, ITC, Johnson & Johnson, Veeba

- Owned architecture and delivery of the WhatsApp **order-agent platform**: decomposed one opaque LangGraph block into seven testable steps (extract, catalog search, alias lookup, LLM match, quantities, cart, reply) composed on a visual canvas via a new “**Zo Flow**” block with per-outcome connectors, so per-customer behaviour changes need no Python edits or redeploy; running in production for Veeba, Noida Bakers and field-ops workspaces
- Designed persistent **product-alias memory** (customer-scoped, word-order-insensitive, fed to the matching LLM as evidence rather than a shortcut) and built a rarity-weighted alias **matcher** (initials, prefixes, typos) over 15K production aliases; real rep phrasings matched to the right product first time rose from **81% to 95%** (48 to 56 of 59)
- Built the order-agent **evaluation system**: 6 eval sets and 64 scenarios graded line-by-line against ground-truth SKUs (no LLM judge), run twice daily in GitHub Actions with a breach rule (score below 90%, errored item, unfinished run) that alerts Teams and feeds a hosted dashboard
- Built **OpenTelemetry-based Langfuse tracing** (span per flow block, traces grouped by conversation, fail-open) and per-step input/output capture rendered in the AI-playground UI, so a failed order turn is diagnosed on one screen instead of in CloudWatch
- Cut catalog-search latency per order line from **~875 ms to ~228 ms** in production via a slim index projection (identical results), and fixed tenant-filter starvation in kNN knowledge search (OpenSearch pre-filter)
- Made order placement multi-location (one checkout per address, sequenced against a cart API that merges same-variant lines) and routed field-rep WhatsApp-group orders through the new pipeline behind a per-workspace flag, with a fallback that can never double-write a cart
- Built full-stack customer journeys: send-time resolution of dynamic WhatsApp template buttons (custom page, book-contract, reorder) across builder UI, communication-v2 resolver and Next.js routes; secure single-use OTP statement links (KrakenD, Copilot API, URL shortener); migrated the customer web-app from webpack to **Next.js 15** SSR on AWS Amplify
- Stopped a production crash loop in communication-v2 (a shared handler exiting on any unhandled rejection) and coalesced duplicate concurrent channel-member loads; made Hasura-triggered dispatch notifications transition-aware to end duplicate WhatsApp sends
- Owned the frontend of the real-time inbox (React, TypeScript, PubNub and Centrifugo listeners, paginated history) and extended Webpack 5 **Module Federation** remotes across a 19-app Nx monorepo
- Promoted to Lead within the first year; lead a team of five engineers: turn business requirements into technical approaches, review designs and pull requests, and own delivery and quality; hired for 3 roles

Sumeru Software – Senior Developer Aug 2022 – Jan 2023

- Built a role-based CRM from scratch (React, Redux Toolkit, Node.js) with permission-based rendering and modular dashboards; owned kickoff-to-production and shipped the MVP in 4 months with a 4-engineer team

Aerchain – Software Development Engineer 1 (converted from intern) Jul 2019 – Jan 2022

- Built a real-time procurement auction module (optimistic UI with server reconciliation) and barcode-scan inventory management with offline-first PWA support (service workers)

Projects

RoleMiner – LLM-scored job discovery pipeline *Python, FastAPI, scikit-learn, React, Docker*

- Staged funnel over 5 ATS types (rule filter, TF-IDF pre-rank, batched LLM scoring): cheap retrieval narrows thousands of postings to a top-50 batch before any model call, keeping a full run under \$0.002

Education

B.E. (Hons) Mechanical Engineering, BITS Pilani

2016 – 2020